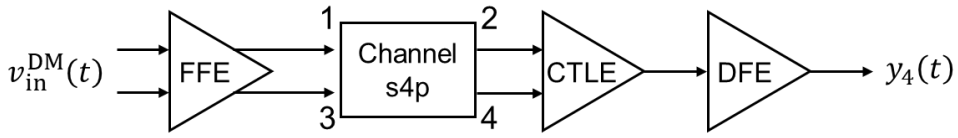
Homework: NRZ vs PAM4

An s4p is provided to you which is a (rather complicated) differential channel, with ports 1&3 at left side and 2&4 at right side. Specifically, the channel consists of a 40-cm differential strip line and a 10-cm of differential microstrip, in total 50-cm routing, and some parasitic LC distributed along the path to model the effects of via, connector, etc. Your task is to conduct experiment on this channel, similar to [A], to see at which bit rate NRZ is better than PAM4, and at which bit rate PAM4 is better.



Constraints

- Assume the source and load ports of the s4p all have impedance 50 Ω (so you only need the S_{dd21}).
- CTLE, FFE, and DFE are required to open the eye. You are free to design their parameters. The equalizers for NRZ and PAM channels need not have identical parameters. The only constraint is that the complexity of the equalizer should be the same or similar, e.g., number of FFE taps, number of DFE taps, number of CTLE poles, and maximum CTLE gain.
- For the NRZ and PAM4 sources, assume they have same bit rate, risetime, and Vpp. They both have sinusoidal jitter with amplitude 1.5 ps and frequency 10 MHz, and Gaussian random jitter with standard deviation σ = 1 ps.

In your report, provide the final FFE+Channel+CTLE+DFE eyes of NRZ and PAM4 at two different bit rates (total four figures), and comment on your results. You also need to provide the equalizer settings and source code. Optional: you may consider showing the step-by-step experiment procedure as in the reference answer below.

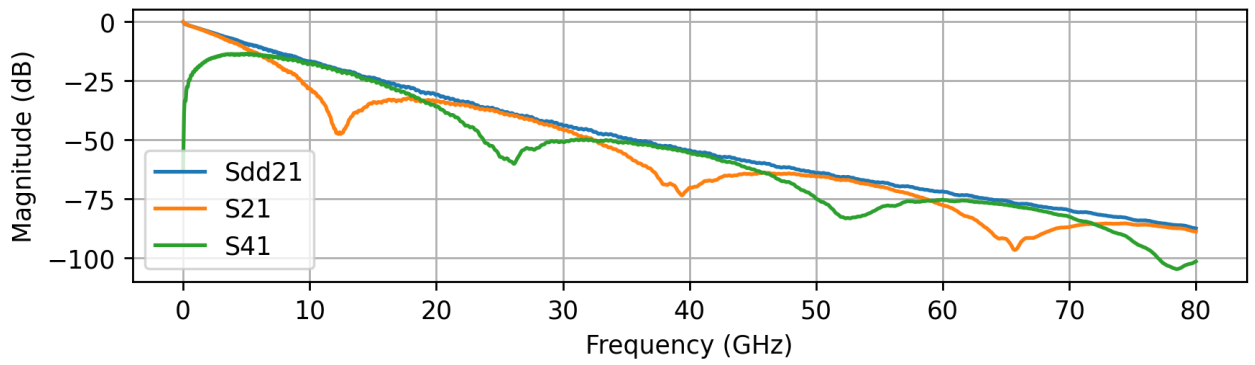
[A] N. Dikhaminjia *et al.*, "High-speed serial link challenges using multi-level signaling," *IEEE EPEPS*, 2015. [online available at researchgate].

Note: in [A], they set SJamp = 0.1UI and RJsigma = 0.05UI. Under such setting, PAM4 would have a larger absolute random jitter and sinusoidal jitter than NRZ, which may not be realistic. Therefore, in this homework, we use the same absolute SJ and RJ for both NRZ and PAM4.

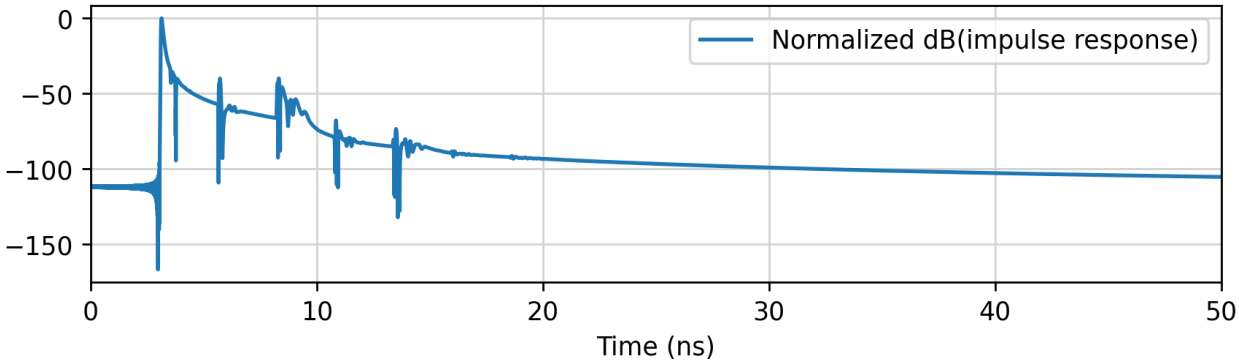
Note: there is no sample code for this homework, because most of the functions are the same as in previous homework except the generation of PAM4 signals and its DFE. You are free to define any function you need.

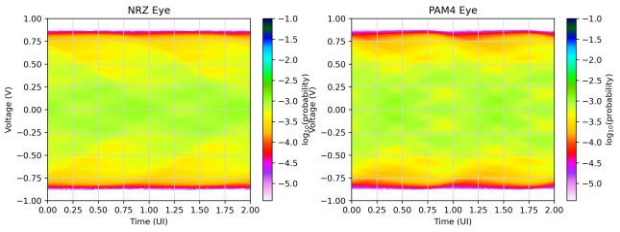
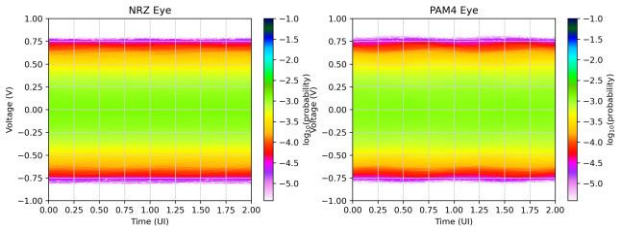
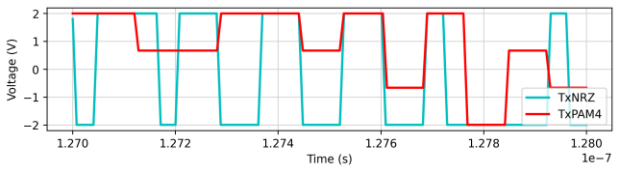
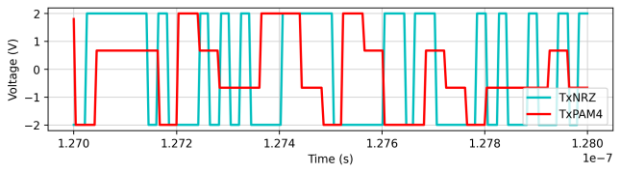
Appendix: Example answer

Check S parameters

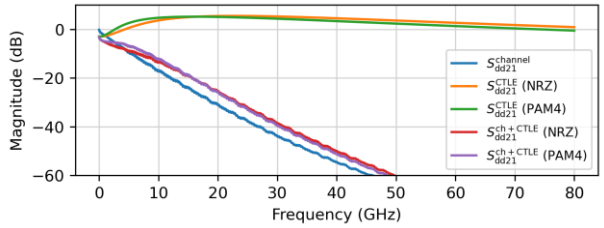


Check impulse response of Sdd21 to determine truncation time

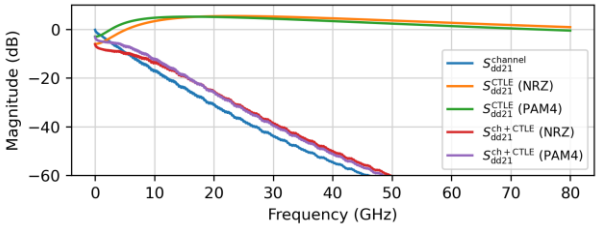


Case 1: Bit rate = X Gbps ⇨ NRZ symbol rate = X Gbaud PAM4 symbol rate = X/2 Gbaud	Case 2: Bit rate = Y Gbps (Y > X) ⇨ NRZ symbol rate = Y Gbaud PAM4 symbol rate = Y/2 Gbaud
Raw channel eyes (no equalizers) 	Raw channel eyes (no equalizers) 
Check input waveform 	Check input waveform 

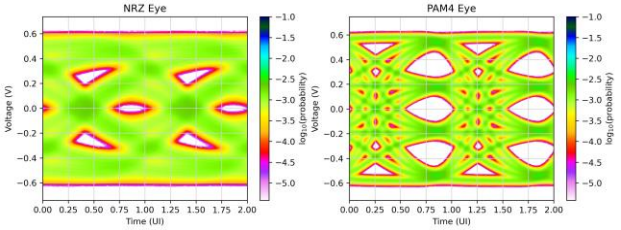
CTLE design



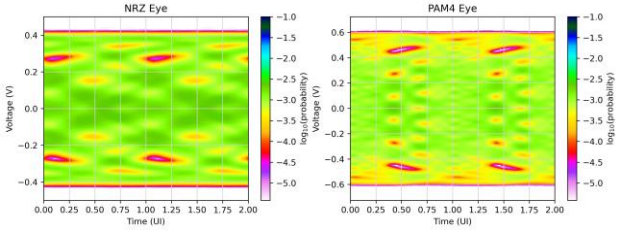
CTLE design



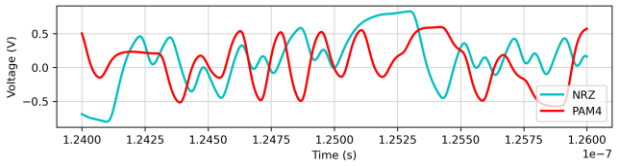
Channel+CTLE eyes



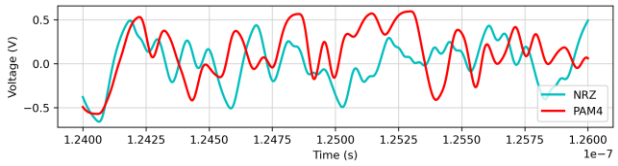
Channel+CTLE eyes



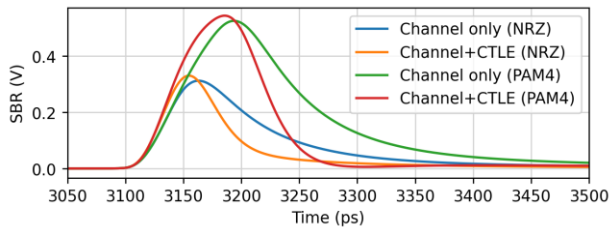
Check time-domain waveform



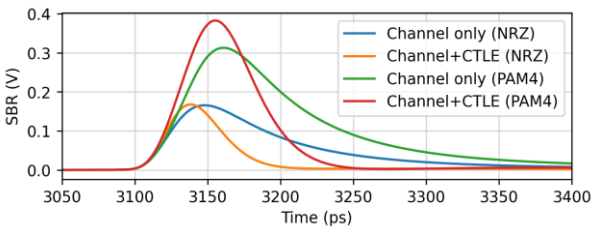
Check time-domain waveform



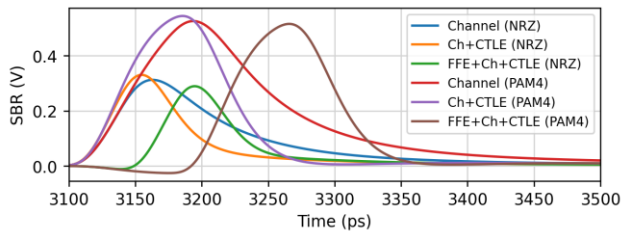
Get SBR for FFE design



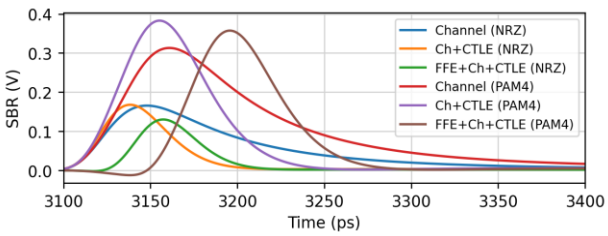
Get SBR for FFE design



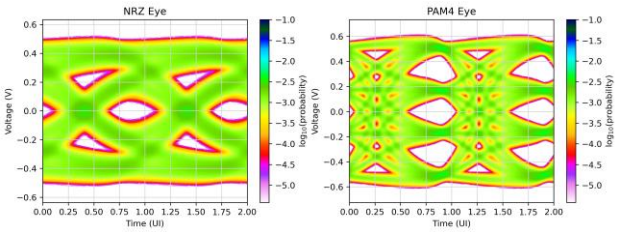
SBR after FFE, used for later DFE design



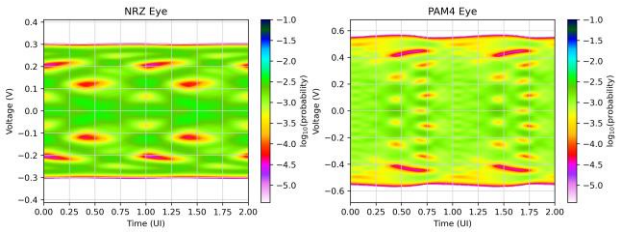
SBR after FFE, used for later DFE design



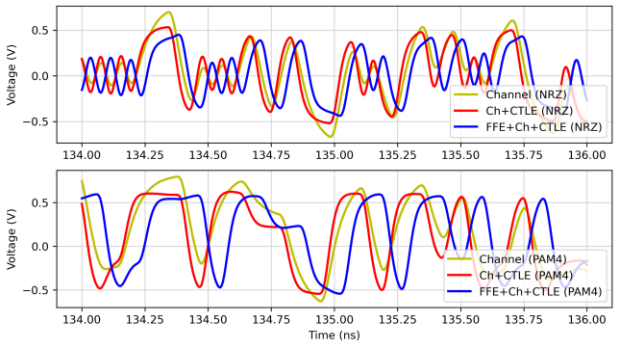
FFE+Channel+CTLE eyes



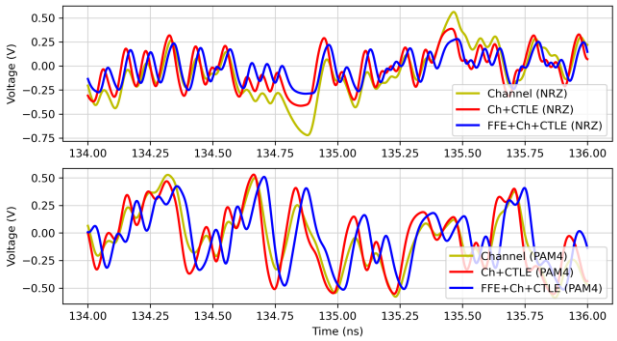
FFE+Channel+CTLE eyes



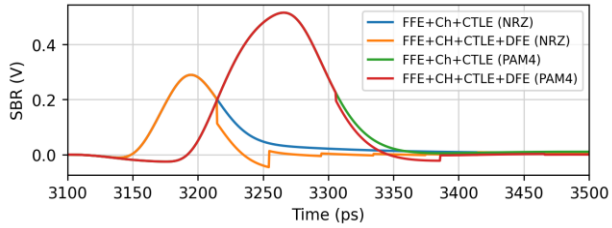
Check time-domain waveform



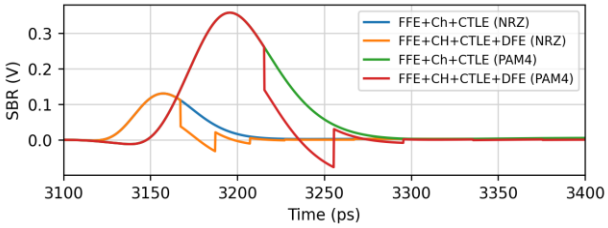
Check time-domain waveform



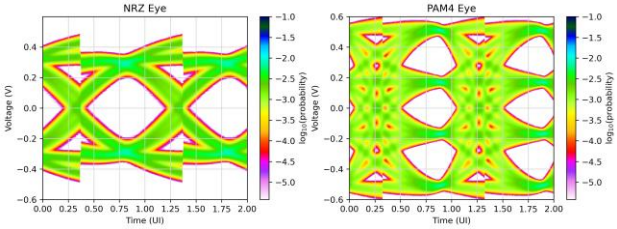
"SBR" after DFE (for visualization only)



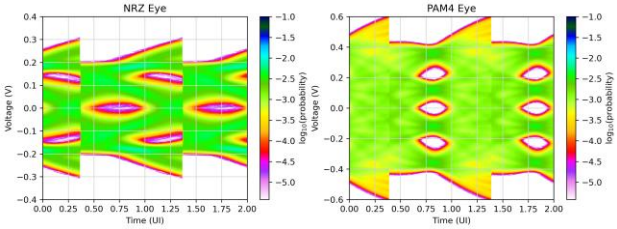
"SBR" after DFE (for visualization only)



FFE+Channel+CTLE+DFE eyes

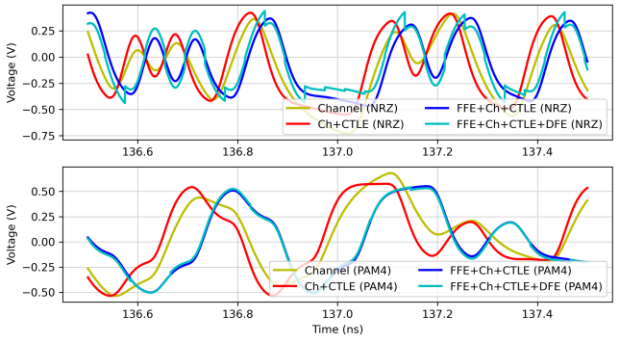


FFE+Channel+CTLE+DFE eyes

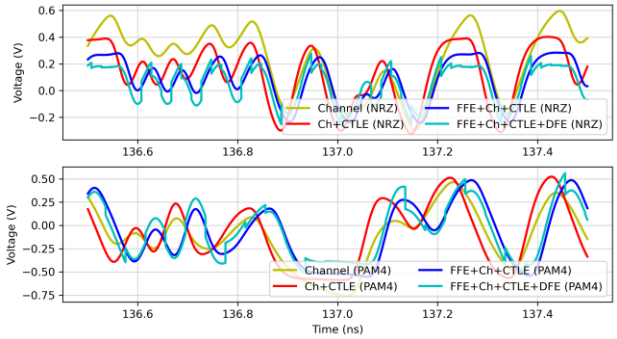


At this step, you may need to adjust the threshold voltages of the PAM4 DFE based on the eye.

Check time-domain waveform



Check time-domain waveform



Observation: at bit rate X, the final NRZ eye has larger EH and EW than the PAM4 eyes, while at bit rate Y, the PAM4 eyes are better.